

Python programming

Where this unit sits, what it covers, and how far

Where this unit sits

No specification reference. The Python course is not an OCR topic and carries no specification reference. It teaches programming in a real language, which Component 02 assesses the thinking behind: a pupil who has written, run and fixed their own programs reads an exam question about iteration differently from one who has only read about it. Where it supports a topic, that topic is named. Where it goes beyond the specification - the editor, the test-based marking, SQL through sqlite3 - it is named as enrichment, not as coverage.

Before it: 1.1 Systems architecture, 1.2 Memory and storage, 1.3 Computer networks, connections and protocols, 1.4 Network security, 1.5 Systems software, 1.6 Ethical, legal, cultural and environmental impacts of digital technology, 2.1 Algorithms, 2.2 Programming fundamentals, 2.3 Producing robust programs, 2.4 Boolean logic, 2.5 Programming languages and Integrated Development Environments

After it: nothing — this is the last unit of the course

The lesson sequence and the questions it answers

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
1	Printing and variables pr-101	Why must a program decide the data type of a value before it can do anything useful with it?	56	108
2	Arithmetic pr-102	How do you choose between /, // and % when the answer to a calculation has to come out as a whole number?	54	104
3	Selection pr-103	Why does the order the branches are written in change what an if ... elif ... else actually does?	55	110
4	Count-controlled loops pr-104	How do you decide what to put inside range() so that a loop counts exactly the values you want?	55	109
5	Condition-controlled loops pr-105	How do you decide between a for loop and a while loop before you write a single line of code?	55	114
6	String handling pr-106	Why are the characters of a string numbered from 0 rather than from 1?	44	84
7	Lists pr-107	Why is the last item of a list always at index len(list) - 1?	44	93
8	Two-dimensional lists pr-108	Which index comes first when a program reads a cell, and why does swapping the two over give a different answer?	43	89
9	Subprograms pr-109	When should a subprogram return a value rather than print one?	43	87
10	File handling pr-110	Why does the mode a file is opened in matter more than anything else on that line?	43	87
11	Searching and sorting pr-111	Why is a binary search so much faster than a linear search on a long list, and what does using one cost you?	45	88

#	LESSON	THE BIG QUESTION	ACTIVITIES	MARKS
12	Validation and authentication pr-112	Why is refusing bad data better than letting a program stop with an error?	46	90
13	SQL queries pr-113	Which clause of a query decides which fields come back, and which clause decides which records?	44	79

Prior knowledge, vocabulary and skills

Assumed before the unit: the outcomes of 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 2.1, 2.2, 2.3, 2.4, 2.5

Vocabulary introduced: Variable, Constant, Casting, Concatenation, Integer division, Modulus, Exponent, Pseudo-random, Selection, Condition, elif, Nested selection, Count-controlled iteration, range(), Running total, Nested loop, Condition-controlled iteration, Validation loop, Sentinel value, Infinite loop, Index, Slice, Character set, ord() and chr(), Array, Traverse, append() and pop(), Two-dimensional list, Row and column, Cell, Subprogram, Parameter and argument, return, Local and global variable, Append mode, Record, Field, split(), Linear search, Binary search, Pass, Comparison, Validation, Format check, Authentication, SQL, Record and field, WHERE, ORDER BY

What a pupil does, not only reads: change, code, complete, debug, match, order, predict, sort, try

Coverage matrix

Every outcome this unit states, and whether it is taught, practised and assessed **within this unit**. Full means all three; partial means some; not covered means none of the three could be traced.

How this was worked out. The trace is derived by matching the content words of an outcome against the station text, the activities and the exam questions of the unit, not authored by hand. The words each match was made on are in build/coverage.json. It shows where to look; it does not certify that the alignment is right, and a full row has not been checked by a person either.

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know how to print lines of output in the order a task asks for pr-l01-o1	pr-l01	yes	yes	yes	Full
know how to hold a value in a variable or a constant, and how to read one with input() pr-l01-o2	pr-l01	yes	yes	yes	Full
understand why a value that has been typed in must be cast before it can be calculated with pr-l01-o3	pr-l01	yes	yes	yes	Full
know which arithmetic operator each kind of calculation needs pr-l02-o1	pr-l02	yes	yes	yes	Full
understand the difference between /, // and % and what each one hands back pr-l02-o2	pr-l02	yes	yes	yes	Full
know how to round a result, and how to draw a random number from a seed pr-l02-o3	pr-l02	yes	yes	yes	Full
know how to write a condition using the comparison operators pr-l03-o1	pr-l03	yes	yes	yes	Full
understand how if, elif and else choose between several outcomes pr-l03-o2	pr-l03	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know how to join two conditions together with and, or and not pr-l03-o3	pr-l03	yes	yes	yes	Full
know how to write a for loop that repeats a block a set number of times pr-l04-o1	pr-l04	yes	yes	yes	Full
understand how range() decides which values the loop counter takes pr-l04-o2	pr-l04	yes	yes	yes	Full
know how to build a running total inside a loop, and how to put one loop inside another pr-l04-o3	pr-l04	yes	yes	yes	Full
know how to write a while loop and what makes it stop repeating pr-l05-o1	pr-l05	yes	yes	yes	Full
understand how a validation loop keeps asking until the value typed in is acceptable pr-l05-o2	pr-l05	yes	yes	yes	Full
know how a sentinel value ends a loop when the amount of data is not known in advance pr-l05-o3	pr-l05	yes	yes	yes	Full
know how to find the length of a string and the character at a given position pr-l06-o1	pr-l06	yes	yes	yes	Full
understand how a slice takes part of a string, and why its end position is left out pr-l06-o2	pr-l06	yes	yes	yes	Full
know how to change case, search inside text, and move between a character and its code pr-l06-o3	pr-l06	yes	yes	yes	Full
know how to create a list and read one item out of it by its index pr-l07-o1	pr-l07	yes	yes	yes	Full
understand how to traverse a list in order to total, count or compare its items pr-l07-o2	pr-l07	yes	yes	yes	Full
know how to add an item onto a list and how to take one out again pr-l07-o3	pr-l07	yes	yes	yes	Full
know how to build a two-dimensional list as a list of lists and fill it from input pr-l08-o1	pr-l08	yes	yes	yes	Full
understand how a row index and a column index together pick out one cell pr-l08-o2	pr-l08	yes	yes	yes	Full
know how to write nested loops that print a grid and total any row or column of it pr-l08-o3	pr-l08	yes	yes	yes	Full

OUTCOME	LESSON	TAUGHT	PRACTISED	ASSESSED	COVERAGE
know how to define a subprogram with def and how to call it pr-l09-o1	pr-l09	yes	yes	yes	Full
understand how a parameter carries a value into a subprogram and how return carries one back out pr-l09-o2	pr-l09	yes	yes	yes	Full
know how to write a subprogram that uses a variable of its own, and one that changes a shared variable pr-l09-o3	pr-l09	yes	yes	yes	Full
know how to open a text file, read what is in it and close it again pr-l10-o1	pr-l10	yes	yes	yes	Full
understand what write mode does to a file, and how append mode differs from it pr-l10-o2	pr-l10	yes	yes	yes	Full
know how to split one line of a file into its separate fields and use them pr-l10-o3	pr-l10	yes	yes	yes	Full
know how to code a linear search and a binary search, and report the position each one finds pr-l11-o1	pr-l11	yes	yes	yes	Full
understand how a bubble sort, an insertion sort and a merge sort each put a list in order pr-l11-o2	pr-l11	yes	yes	yes	Full
know how to count the comparisons an algorithm makes, in order to compare two of them pr-l11-o3	pr-l11	yes	yes	yes	Full
know how to code a presence check, a type check and a range check pr-l12-o1	pr-l12	yes	yes	yes	Full
understand how a length check and a format check differ, and why the length is tested first pr-l12-o2	pr-l12	yes	yes	yes	Full
know how to write a validation loop, and an authentication check that allows a set number of attempts pr-l12-o3	pr-l12	yes	yes	yes	Full
know how SELECT and FROM choose the fields and the table a query reads pr-l13-o1	pr-l13	yes	yes	yes	Full
understand how WHERE decides which records a query hands back pr-l13-o2	pr-l13	yes	yes	yes	Full
know how to join conditions with AND and OR, and how to put results in order with ORDER BY pr-l13-o3	pr-l13	yes	yes	yes	Full

39 full · 0 partial · 0 not covered, of 39 outcomes.

Gaps, assumptions and overlap

None found by the trace.

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